

Problem set 7

Information Retrieval Problems

Excercise 1

The words car, auto, best should have the following frequencies in the ten documents Doc1, .., Doc10. Simply assume that all other words in the documents are stop words. Compute the tf-idf weight for each term.

term\doc	Doc1	Doc2	Doc3	Doc4	Doc5	Doc6	Doc7	Doc8	Doc9	Doc10
car	3	0	0	5	12	0	0	2	8	1
auto	8	6	0	12	0	0	9	1	3	10
best	0	1	7	0	1	5	12	0	2	0

Figure 1: Term Frequency Table for Each Document.

Solution

Total number of documents: $N = 10$

Step 1: Document Frequency (DF)

- car appears in Docs: $D_1, D_4, D_5, D_8, D_9, D_{10} \Rightarrow df_{\text{car}} = 6$
- auto appears in Docs: $D_1, D_2, D_4, D_7, D_8, D_9, D_{10} \Rightarrow df_{\text{auto}} = 7$
- best appears in Docs: $D_2, D_3, D_5, D_6, D_7, D_9 \Rightarrow df_{\text{best}} = 6$

Step 2: Inverse Document Frequency (IDF)

Using $\log_{10}$:

$$idf_t = \log_{10} \left(\frac{N}{df_t} \right)$$
$$idf_{\text{car}} = \log_{10} \left(\frac{10}{6} \right) = 0.222$$

$$idf_{\text{auto}} = \log_{10} \left(\frac{10}{7} \right) = 0.155$$

$$idf_{\text{best}} = \log_{10} \left(\frac{10}{6} \right) = 0.222$$

Step 3: TF-IDF Calculation

$$tfidf_{t,d} = tf_{t,d} \times idf_t$$

Final TF-IDF Table

Doc	car	auto	best
D_1	$3(0.222) = 0.666$	$8(0.155) = 1.240$	0
D_2	0	$6(0.155) = 0.930$	$1(0.222) = 0.222$
D_3	0	0	$7(0.222) = 1.554$
D_4	$5(0.222) = 1.110$	$12(0.155) = 1.860$	0
D_5	$12(0.222) = 2.664$	0	$1(0.222) = 0.222$
D_6	0	0	$5(0.222) = 1.110$
D_7	0	$9(0.155) = 1.395$	$12(0.222) = 2.664$
D_8	$2(0.222) = 0.444$	$1(0.155) = 0.155$	0
D_9	$8(0.222) = 1.776$	$3(0.155) = 0.465$	$2(0.222) = 0.444$
D_{10}	$1(0.222) = 0.222$	$10(0.155) = 1.550$	0

Final Answer

- $idf_{\text{car}} = 0.222$, $idf_{\text{auto}} = 0.155$, $idf_{\text{best}} = 0.222$
- TF-IDF values are computed as $tf \times idf$ for each document as shown above

Exercise 2

Assume the following documents. Assume that the following terms are stop words: is, and, in, the, was, each, other. Construct an inverted index.

Assume the following documents

doc-id	text
Doc1	"italy is world champion 2006"
Doc2	"germany and italy played each other in the semifinal"
Doc3	"germany was in the semifinal 2006"
Doc4	"germany won the semifinal in italy 1990"

Figure 2: Documents in a corpus.

Solution: Inverted Index Construction

Given Documents

- Doc 1: The cat is in the hat
- Doc 2: The cat and the dog
- Doc 3: The dog was in the fog
- Doc 4: Each cat and each dog

Stop Words

is, and, in, the, was, each, other

Step 1: Remove Stop Words

- Doc 1 $\rightarrow$ cat, hat
- Doc 2 $\rightarrow$ cat, dog
- Doc 3 $\rightarrow$ dog, fog
- Doc 4 $\rightarrow$ cat, dog

Step 2: Construct Inverted Index

- cat $\rightarrow$ {Doc 1, Doc 2, Doc 4}
- dog $\rightarrow$ {Doc 2, Doc 3, Doc 4}
- hat $\rightarrow$ {Doc 1}
- fog $\rightarrow$ {Doc 3}

Final Inverted Index

cat $\rightarrow$ { $D1, D2, D4$ }

dog $\rightarrow$ { $D2, D3, D4$ }

hat $\rightarrow$ { $D1$ }

fog $\rightarrow$ { $D3$ }

Excercise 3

Consider following documents in the corpus.

- Doc 1 breakthrough drug for schizophrenia.
- Doc 2 new schizophrenia drug.
- Doc 3 new approach for treatment of schizophrenia.

Assume the following documents

doc-id	text
Doc1	"italy is world champion 2006"
Doc2	"germany and italy played each other in the semifinal"
Doc3	"germany was in the semifinal 2006"
Doc4	"germany won the semifinal in italy 1990"

Figure 3: Documents in a corpus.

- Doc 4 new hopes for schizophrenia patients.

What are the returned results for these queries:–

- schizophrenia AND drug
- schizophrenia AND NOT drug
- schizophrenia OR drug

Solution

Given documents:

- Doc 1: breakthrough drug for schizophrenia
- Doc 2: new schizophrenia drug
- Doc 3: new approach for treatment of schizophrenia
- Doc 4: new hopes for schizophrenia patients

1. Query: schizophrenia AND drug

This query retrieves documents containing **both** terms “schizophrenia” and “drug”.

- Doc 1: contains both ✓
- Doc 2: contains both ✓
- Doc 3: contains schizophrenia only ×
- Doc 4: contains schizophrenia only ×

Result: {Doc 1, Doc 2}

2. Query: schizophrenia AND NOT drug

This query retrieves documents that contain “schizophrenia” but **do not contain** “drug”.

- Doc 1: contains drug ×
- Doc 2: contains drug ×
- Doc 3: no drug ✓
- Doc 4: no drug ✓

Result: {Doc 3, Doc 4}

3. Query: schizophrenia OR drug

This query retrieves documents that contain **either** “schizophrenia” or “drug” (or both).

- Doc 1: contains both ✓
- Doc 2: contains both ✓
- Doc 3: contains schizophrenia ✓
- Doc 4: contains schizophrenia ✓

Result: {Doc 1, Doc 2, Doc 3, Doc 4}

Final Answers

- schizophrenia AND drug $\rightarrow$ {Doc 1, Doc 2}
- schizophrenia AND NOT drug $\rightarrow$ {Doc 3, Doc 4}
- schizophrenia OR drug $\rightarrow$ {Doc 1, Doc 2, Doc 3, Doc 4}

Exercise 4

Recommend a query processing order for

(tangerine OR trees) AND (marmalade OR skies) AND (kaleidoscope OR eyes)

given the following postings list sizes:

Solution To minimize the cost of Boolean query processing, we should:

1. process the **OR** subqueries first inside each parenthesized group,
2. then intersect the resulting intermediate lists in increasing order of their estimated sizes.

Term	Postings size
eyes	213312
kaleidoscope	87009
marmalade	107913
skies	271658
tangerine	46653
trees	316812

Table 1: Terms and the correponsing posting list size

The query is:

(tangerine OR trees) AND (marmalade OR skies) AND (kaleidoscope OR eyes)

Using the postings sizes:

$$|\text{tangerine}| = 46653, \quad |\text{trees}| = 316812$$

$$|\text{marmalade}| = 107913, \quad |\text{skies}| = 271658$$

$$|\text{kaleidoscope}| = 87009, \quad |\text{eyes}| = 213312$$

The sizes of the OR groups are at most the sums of the postings sizes:

$$|\text{tangerine} \cup \text{trees}| \leq 46653 + 316812 = 363465$$

$$|\text{marmalade} \cup \text{skies}| \leq 107913 + 271658 = 379571$$

$$|\text{kaleidoscope} \cup \text{eyes}| \leq 87009 + 213312 = 300321$$

So the smallest intermediate result is expected to be:

(kaleidoscope OR eyes)

followed by

(tangerine OR trees)

followed by

(marmalade OR skies)

Therefore, a good query processing order is:

((kaleidoscope OR eyes) AND (tangerine OR trees)) AND (marmalade OR skies)

Equivalently, in words:

1. compute (kaleidoscope OR eyes),
2. compute (tangerine OR trees),
3. intersect those two results,
4. compute (marmalade OR skies),
5. intersect that final group with the previous result.

Hence, the recommended processing order is:

(kaleidoscope OR eyes) $\rightarrow$ (tangerine OR trees) $\rightarrow$ AND $\rightarrow$ (marmalade OR skies) $\rightarrow$ AND.

Exercise 5

In constructing the index, which step is most expensive/complex?

Solution The most computationally expensive step in building an inverted index is the sorting and merging of term-document pairs.

Exercise 6

Why don't we use a relational database for information retrieval?

Solution We don't use relational databases for information retrieval because they are designed for exact, structured queries, whereas IR requires efficient full-text search and relevance-based ranking over large unstructured document collections.

Exercise 7

In the BM25/Okapi formula, which term is the 'inverse document frequency'? Which term is the 'term frequency'? Where is the length normalization?

Solution The term frequency (with length normalization) is:

$$\frac{(K_1 + 1) tf}{K_1 \left((1 - b) + b \frac{dl}{avg(dl)} \right) + tf}$$

The inverse document frequency is:

$$\ln \frac{N - df + 0.5}{df + 0.5}$$

The length normalization is:

$$K_1 \left((1 - b) + b \frac{dl}{avg(dl)} \right)$$

Exercise 8

Given the following corpus:

- D1 I love cats cats cats.
- D2 I love cats and I love dogs.
- D3 I love dogs.
- D4 I dont like cats.
- D5 I love puppies.
- D6 I love kittens.

And the following queries:

- **Q1:** cats
- **Q2:** dogs

First, start with standard assumptions: a large corpus ($N=1000$), few relevant documents ($S=3$), and each term occurs in only the documents above. Because the numbers are small relative to N , you can assume that $N-n$ or $N-S$ is approximately N . Also assume $p_i=0.5$; i.e., if a term is relevant, it has a 0.5 probability of occurring in a relevant document. Calculate the Retrieval Status Value for each query and the documents above, and rank the documents.

Solution

$$\text{RSV}(Q1, D1) = \sum_{x_i=q_i=1} \log \frac{p_i(1-r_i)}{r_i(1-p_i)} \approx \sum_{x_i=q_i=1} \log \frac{N}{n_i} = \log \frac{1000}{3}$$

D2 and D4 are the same since cats appears in them). D3, D5 and D6 are 0, as the sum includes no items (nothing for which $x_i = q_i = 1$).

This gives a ranking of $D1 = D2 = D4 > D3 = D5 = D6$.

The only difference is in $n_i = 2$ (and which queries have the term), so:

$$\text{RSV}(Q2, D2) = \sum_{x_i=q_i=1} \log \frac{p_i(1-r_i)}{r_i(1-p_i)} \approx \sum_{x_i=q_i=1} \log \frac{N}{n_i} = \log \frac{1000}{2}$$

D3 is the same. D1, D4, D5 and D6 are 0, as the sum includes no items (nothing for which $x_i = q_i = 1$).

This gives a ranking of $D2 = D4 > D1 = D3 = D5 = D6$.

Exercise 9

How does the size of the index impact search performance? Does a larger index size guarantee a better performance? Why or why not? (Remember, search performance is asking how relevant the returned documents are, not how fast it runs.)

Solution

- The search engine needs to have enough and diverse document collection to serve diverse users information needs. However, the larger corpus also poses challenges to the ranking models, as they now need to deal with more documents. After all, the search system can show a fixed number of documents at the top (e.g., 10) despite having a larger index size.
- The search performance (i.e., relevance) depends on many factors; thus, increasing the index size does not guarantee better performance. Some good index choices can be to avoid duplication, diversify corpus, and cover as much topic as possible based on the expected users type.

Excercise 10

What is the idf of a term that occurs in every document?

Solution 0